

Evaluating Avoidable Heterogeneity in Exploratory Factor Analysis Results

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Abstract

Meaningful interpretations of scores derived from psychological scales depend on the replicability of psychometric properties. Despite this, and unexpected inconsistencies in psychometric results across studies, psychometrics has often been overlooked in the replication literature. In this article, we begin to address replication issues in exploratory factor analysis (EFA). We use a Monte Carlo simulation to investigate methodological choices made throughout the EFA process that have the potential to add heterogeneity to results. Our findings show that critical decision points for EFA include the method for determining the number of factors as well as rotation. The results also demonstrate the relevancy of data characteristics, as some contexts are more susceptible to the effects of methodological choice on the heterogeneity of results.

Translational Abstract

When using psychological scales, such as a survey or questionnaire that is intended to measure depression, for example, meaningful interpretations of (depression) scores depend on the replicability of results from psychometric studies. Despite this, and despite unexpected inconsistencies in psychometric results across studies in the empirical literature, psychometrics has often been overlooked in the replication discussion. In this article, we begin to address replication issues in exploratory factor analysis (EFA)—a popular method for scale development and evaluation. We use a Monte Carlo simulation to investigate whether choices made throughout the data analysis process are potentially detrimental to the replicability of EFA results. Our findings show that critical decision points for EFA include the method for determining the number of factors as well as rotation. The results also show that the characteristics of the data are an important consideration as some contexts have a greater impact on efforts to replicate an EFA.

Keywords: exploratory factor analysis, replication, methodological flexibility, researcher degrees of freedom, questionable research practices

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In the last two decades, psychologists have placed growing emphasis on the replication of empirical findings. As the discussion surrounding replication has evolved, the methodological focus has shifted away from binary assessments of whether something has replicated toward addressing more nuanced aspects of replication, including considering different goals for replication (e.g., Anderson & Maxwell, 2016; Hedges & Schauer, 2019a; Steiner et al., 2019), applying multistudy approaches to evaluate replication

more cumulatively (e.g., Hedges & Schauer, 2019b), and improving the replicability of original studies (e.g., Nosek et al., 2015; Silberzahn et al., 2018). However, with the exception of a handful of studies (e.g., Flake et al., 2017, 2022; Flake & Fried, 2020; Hussey & Hughes, 2020; Shaw et al., 2020), psychometrics remains largely absent from the replication discussion, which is unfortunate given the important role that psychometrics plays in many scientific endeavors. If the psychometric properties of a scale/score (e.g., coefficient α , factor structure) fail to replicate in conditions where they would be expected to, inferences made using scores from that scale can be questionable and/or uninterpretable (Flake et al., 2017, 2022). In this study, we consider the replicability of psychometric properties in factor analysis—a widely used method for scale development and evaluation.

Although this is a relatively new area of methodological focus, there is empirical evidence in the factor analytic literature that can be considered issues of replication. For example, consider the Self-Compassion Scale (SCS; Neff, 2003). Despite wide use, adaptations, and translations (e.g., Cleare et al., 2018; Kumlander et al., 2018; Neff et al., 2017, 2019), researchers have noted inconsistencies in factor structure across studies (e.g., Cleare et al., 2018; Kumlander et al., 2018). Evidence is mixed regarding whether the common six-factor conceptualization of the SCS is supported or not, even for studies involving the same population (e.g., Cleare et al., 2018; Williams et al., 2014). Similar issues have been raised

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for the Positive and Negative Affect Schedule (PANAS) as well as the Five Facets Mindfulness Questionnaire (FFMQ). Regarding the PANAS, Seib-Pfeifer et al. (2017) attempted to replicate four predominant PANAS models from prior research (e.g., uncorrelated two-factor model, correlated two-factor model, three-factor model splitting up the Negative Affect factor, and bifactor model including an Affective Polarity factor). Although the bifactor model achieved the best fit, the results were ultimately inconclusive. Regarding the FFMQ, Lecuona et al. (2020) were unsuccessful in conceptually replicating any of the 12 key factor structures identified based on 36 prior studies.

Personality instruments are also commonly at the center of factor structure debates. For example, Cattell's theory of 16 personality factors (16PF; Cattell, 1957) has been criticized for its failure to replicate. Howarth and Browne (1971) were unable to replicate Cattell's theory and concluded the 16PF did not measure the factors it was supposed to measure. Additionally, Noller et al. (1987) stated that there was a substantial amount of overlap between the 16PF and that fewer factors would be more appropriate.

More generally, these inconsistencies and unsuccessful replication attempts reflect heterogeneity across study results, which can make it more difficult to accurately infer whether studies agree, particularly when more heterogeneity is present than expected given differences in populations and settings. We contend that some of this heterogeneity is due to the influence of different methodological choices made by researchers during psychometric analysis. However, despite literature identifying areas of methodological flexibility in standard parametric analysis (e.g., researcher degrees of freedom; Simmons et al., 2011, questionable research practices; John et al., 2012), there remains a need to identify and examine the influence of this flexibility in psychometrics. Moreover, further insight into these empirical findings on factor analysis could be gained from the level of control offered by Monte Carlo simulations. We sought to extend the empirical evaluation of factor analytic replication and, using a simulation, consider the role of methodological flexibility within EFA.

The reader may wonder why we are evaluating replication within the context of exploratory analysis, given that replication is often explicitly tied to confirmatory settings (i.e., confirmatory factor analysis [CFA]). There are two answers to this question. First, the name "exploratory" factor analysis is somewhat unfortunate, as it is simply a less restricted version of the common factor model used in CFA applications. Thus, although commonly known as EFA, it is not as "exploratory" as the methods that commonly fall under the umbrella of exploratory techniques. Furthermore, techniques such as target rotation (Browne, 2001) can make EFA more confirmatory in nature. Second, and more importantly, the way in which EFA is used in the psychometric context, to help determine the underlying structure, would be entirely undermined if the results did not replicate. Thus, EFA's value to the psychometric enterprise, whether focusing on EFA itself or EFA as a prelude to CFA, depends heavily on the replicability of the results in situations where replication should be expected. In other words, even though EFA results are expected to vary to some degree from sample to sample, a "practical equivalence" in the number of factors, factor structure, and/or factor labels is crucial for the method to add value to scale development and evaluation.

When considering heterogeneity across results, we use the "practical equivalence" referred to in the previous paragraph to broadly

define replication for our purposes. The notion of replication as "practical equivalence" holds that results across studies are "almost the same" (Hedges & Schauer, 2019b, p. 559). Although this is one of many existing operationalizations of replication (see Fabrigar & Wegener, 2016; Schmidt, 2009), we felt it was a reasonable choice because this reflects how factor analytic results tend to be interpreted empirically and do not carry the unrealistic constraint that EFA results must be identical. Specifically, we focus on the following types of practical equivalence (i.e., factor analytic replication; Manapat et al., 2023): (a) replication of the number of factors and (b) replication of the factor structure. Replication of these aspects of the factor model is crucial for building a strong validity case (e.g., Flake et al., 2017).

In the following sections, we first discuss the broader issue of heterogeneity in research results, describing how heterogeneity relates to replication and how methodological flexibility is a possible source of heterogeneity. We then discuss an evaluation of factor analytic replication in the empirical literature and use this work to guide a simulation we conducted to assess the impact of methodological flexibility on the replication of EFA results. Specifically, we examine how different methodological options in EFA affect the number of factors as well as multiple aspects of the factor solution (e.g., factor loadings and factor correlations).

Considerations for Psychometric Replication

Heterogeneity in Research Results

Across studies testing similar phenomena, differences in results should be expected to at least some degree, reflecting sampling error as well as differences in populations, contexts, settings, and times (Kenny & Judd, 2019; McShane et al., 2019; McShane & Böckenholt, 2014). Although heterogeneity is a fundamental part of scientific observation, additional sources of heterogeneity can complicate the interpretation of results. For example, in a study where 29 research teams analyzed the same data set, probed the same research question, and had their analytic strategies peer-reviewed by other research teams, Silberzahn et al. (2018) found effects (in odds-ratio units) to range from 0.89 (slightly negative) to 2.93 (moderately positive), which was an unexpected amount of heterogeneity given such a highly controlled setting. To more clearly describe the sources of heterogeneity in replication studies and the literature more broadly, researchers have made a theoretical distinction between unavoidable and avoidable sources of heterogeneity (e.g., Manapat et al., 2022). Unavoidable heterogeneity is expected, originating from sources such as sampling variability (when the selected sample sizes are appropriately planned) or differences in settings, populations, or conditions. On the other hand, (potentially) avoidable heterogeneity originates from unnecessary sources and can lead to a seeming lack of replication and/or inaccurate evaluations of replication.

Methodological Flexibility: A Source of Heterogeneity

One source of avoidable heterogeneity is methodological flexibility in research and statistical practices. For most statistical analyses, there are many methodological options to consider. Choosing different methodological options, even defensible ones, has the potential to produce different statistical results (e.g., Silberzahn et al., 2018). Given that heterogeneity can result from the use of different

defensible practices, this is a good place to note that we use the term “avoidable” with reference to an ideal world. In other words, not all sources of avoidable heterogeneity can (or should) be immediately removed, because methodologists have rarely agreed upon a single correct way to conduct analyses. However, some of this avoidable heterogeneity can be reduced by eliminating indefensible choices and identifying areas where the decision-making process can be further streamlined. Our main message is that, considering the “garden of forking paths” (Gelman & Loken, 2014), a greater number of subjective decisions increases the potential for heterogeneity across results (Silberzahn et al., 2018).

To understand the ways in which methodological flexibility can result in avoidable heterogeneity, it is helpful to clarify what is meant by methodological flexibility in the first place. Readers may be familiar with various terms used to describe methodological flexibility, including researcher degrees of freedom and questionable research practices. In the present study, we invoke the taxonomy developed by Manapat et al. (2022) to more clearly identify practices that contribute to avoidable heterogeneity in EFA. We briefly describe this taxonomy here. Evaluations are based on the weight of evidence for or against the use of a research practice. The taxonomy first separates responsible conduct of research (best practices) from questionable research practices (unacceptable practices that are not empirically supported). Under responsible conduct of research, a practice is either a single best option or a researcher degree of freedom. Although rare, single best options are optimal in all situations and hence the only acceptable choice. In the many cases where there is no single best option, then the set of empirically supported practices are researcher degrees of freedom (and hence acceptable practices). By more clearly delineating acceptable and unacceptable practices, the taxonomy makes it easier to investigate the complex series of nested decisions made within EFA.¹

Now that we have further defined methodological flexibility, we consider the multiple ways in which this can add avoidable heterogeneity across results in a hypothetical research literature. First, it is important to differentiate between heterogeneity in *decisions* and heterogeneity in *results*. Heterogeneity in decisions regarding what methods and practices are adopted across studies may confer heterogeneity across the results of those studies, and more available options may increase the resulting heterogeneity. We do not claim that questionable research practices are necessarily always worse for heterogeneity compared to researcher degrees of freedom. For example, if all researchers adopt a particular questionable research practice, results may be biased but could still be relatively homogeneous. However, when a subset of researchers selects questionable research practices (via heterogeneity in decisions), the inclusion of their potentially biased results into the literature has the potential to exacerbate heterogeneity in results (Anderson & Liu, 2023).

Second, in addition to *between-decision* heterogeneity (heterogeneity in results among researchers who choose different methodological options), we highlight that there may be *within-decision* heterogeneity (heterogeneity in results among researchers who choose the same option). Some methodological options have a wider distribution in how they behave across different uses, and it is helpful to identify these high-variability options, which may further contribute to heterogeneity. Overall, although there will always be some degree of ambiguity or flexibility in research practices, it can be reduced by limiting questionable research practices, particularly options that have variable performance.

Factor Analytic Replication in the Empirical Literature

Now that we have established a conceptual link between methodological flexibility and heterogeneity in results, we review a study that focused on methodological flexibility in factor analysis specifically. In an empirical evaluation conducted by Manapat et al. (2023), heterogeneity in factor analytic results was observed across studies for three different scales (Brief Self-Control Scale, Center for Epidemiologic Studies-Depression Scale, and NEO Personality Inventory-Revised). This was the case for even the most basic level of factor analytic replication—the number of factors. Using the same research practices taxonomy that we use here, the authors also identified researcher degrees of freedom and questionable research practices in factor analysis that may be contributing to the observed heterogeneity. These findings helped guide our choices for which methodological options to manipulate in the current simulation study. We focused on widely used and/or recommended practices and acknowledge that our list below is not inclusive of every methodological practice possible in EFA. In the following section, we describe these practices and the rationale behind each researcher degree of freedom-questionable research practice distinction.

Researcher Degrees of Freedom in Factor Analysis

The first set of researcher degrees of freedom we consider in the present study is the choice between retaining the full scale or creating a shortened version. In the sample of factor analytic studies reviewed by Manapat et al. (2023), 43% dropped items from the original scale (12 studies out of the 28 that reported a final version of the scale). The most common reason for dropping items was that factor loadings fell below an arbitrary cutoff. Other reasons for dropping items included cross-loadings, reverse-coding, and conforming to a previous study. In many cases, no justification was explicitly provided for dropping items from a scale during EFA.

The second researcher degree of freedom we consider is choosing between two different versions of parallel analysis (PA; Horn, 1965). PA remains one of the most widely used and recommended approaches for determining the number of factors (e.g., Auerswald & Moshagen, 2019; Fabrigar et al., 1999; Garrido et al., 2013). More detail on the difference between the two variations is provided in the Method section. Another popular rule-of-thumb for determining the number of factors is the scree test (Cattell, 1966). Although worth mentioning, we did not include the scree test in the current simulation due to its subjective nature.²

The final set of researcher degrees of freedom we consider is different EFA rotations. Oblique quartimin and oblique promax are widely used, recommended in the factor analysis literature (see Browne, 2001

¹ A reviewer raised an interesting point about the possibility of “agreed upon options,” which may not be methodologically optimal but are commonly applied within a substantive research area. This is an important consideration especially in cases where methodological consensus and consensus among substantive researchers do not align (e.g., the agreed upon option is a questionable research practice). Ideally, the agreed upon option is a single best option or researcher degree of freedom.

² The scree test relies on a scree plot where eigenvalues are plotted against successive factors. Determining the number of factors is based on the “elbow” or the point where there is a sudden drop in the magnitude of the eigenvalues. However, elbows are not always distinct, and decisions about what constitutes the elbow can be subjective (see DeVellis, 2017).

for quartimin; Floyd & Widaman, 1995 for promax; Fabrigar et al., 1999 for both quartimin and promax), and available in commonly used software for EFA. For example, quartimin and promax are available in the *fa* function (*psych* package; Revelle, 2019) in R (R Core Team, 2020) and *Mplus* (specifying CF-QUARTIMAX for direct quartimin; Muthén & Muthén, 2017). Additionally, promax is available in SPSS (IBM Corp, 2020) and quartimin is available in CEFA (specifying CF-QUARTIMAX for direct quartimin; Browne et al., 2010).

Questionable Research Practices in Factor Analysis

With respect to questionable research practices, we consider three practices observed by Manapat et al. (2023). The first is using the Kaiser–Guttman criterion/eigenvalue-greater-than-one rule (K1; Kaiser, 1960) to determine the number of factors. Despite being discouraged (e.g., Fabrigar et al., 1999), K1 is still widely used (e.g., Garrido et al., 2013) and the default in some statistical software (e.g., SPSS).

The second questionable research practice we consider is the use of principal components³ analysis (PCA) for EFA purposes. The goal of PCA is to reduce a large set of observed variables to a smaller set of composite variables (components) that maximize the variance accounted for by the original set of variables. Given this, “components in PCA are conceptually and mathematically quite different from factors in EFA” (Preacher & MacCallum, 2003, p. 20). Yet, components from PCA are often interpreted as latent factors and in some software, PCA is the default approach for factor analysis (e.g., SPSS). For more details on the differences between PCA and EFA, see Byrne (2005), Fabrigar et al. (1999), Preacher and MacCallum (2003), and Widaman (2018).

The last questionable research practice we consider is using orthogonal rotations absent any theoretical justification. In general, constraining factor correlations to zero with orthogonal rotations is rarely warranted in psychological research (e.g., Byrne, 2005; Preacher & MacCallum, 2003) and is known to lead to less desirable solutions (e.g., Fabrigar et al., 1999).

The Current Study

In this study, we considered the impact of various methodological choices on the heterogeneity of EFA results. Our goal was to highlight decision points that have a large impact on results, which may increase the potential of avoidable heterogeneity across a series of EFAs. For example, if a methodological option with highly variable answers is used, there is greater potential to push researchers down more heterogeneous paths and toward more heterogeneous results. We were also interested in the extent to which the impact of these decision points depended on the features of the data.

Method

We conducted a Monte Carlo simulation study to examine the impact of methodological flexibility on EFA results. All data generation and analyses were conducted in R (R Core Team, 2020), and R scripts used for the simulation have been made publicly available on GitHub (Manapat, 2023; <https://github.com/patrickmanapat/efaHet>). We manipulated the following features of the generating factor model: sample size, the number of factors, the number of indicators

per factor, the magnitude of factor correlations, and the magnitude of factor loadings.

Consistent with previous EFA work by MacCallum et al. (1999, 2001), we set sample sizes to $N = 60$, $N = 100$, $N = 200$, and $N = 400$. We generated data from a one-, three-, or five-factor model with eight or 12 indicators per factor. These values for number of factors and indicators per factor are common for widely used scales in psychological research (Manapat et al., 2023). To represent low, medium, and high levels of association, we set generating factor correlations (φ) to 0.3, 0.5, and 0.7, respectively. Generating factor loadings (λ) were set to 0.45, 0.71, and 0.89. These values are consistent with the lowest, middle, and highest communality from the “wide” range of communalities used in the MacCallum et al. (1999, 2001) simulations. We note that the magnitude of generating factor correlations and factor loadings were constant within a condition. For example, if a condition had three factors, eight indicators per factor, factor correlations of 0.3, and factor loadings of 0.45, all three factor correlations would be 0.3 and all 24 factor loadings would be 0.45 in the generating model. Each indicator (j) loaded onto only one factor (i.e., independent clustering). We computed uniquenesses (u_{ij}) for the i th simulee on the j th indicator so that the resulting unique variance would equal: $\sqrt{1 - \lambda^2}$.

Excluding the manipulation of factor correlations for generating models with one factor, all levels were crossed creating a total of 168 conditions, and we generated 5,000 data sets for each condition. First, we drew factor scores (η_i) from a multivariate normal distribution defined by a vector of means (μ) equal to zero and correlation matrix (Σ) with all off-diagonal elements equal to the same value ($\varphi = 0.3$, $\varphi = 0.5$, or $\varphi = 0.7$). Then, we generated data (x_{ij}) according to the following factor model:

$$x_{ij} = \lambda\eta_i + u_{ij} \quad (1)$$

A (Simulated) Garden of Forking Paths

To simulate a garden of forking paths, we analyzed each unique data set in different ways. The first decision point we simulated was the method used to determine the number of factors. This included K1 and two approaches for PA. To implement K1, we used the *eigen* function (base R) to compute eigenvalues, and the number of eigenvalues greater than one suggested how many factors should be retained. For the two PA methods, we used the *fa.parallel* function in the *psych* package (Revelle, 2019). The number of iterations was set to 100 ($n.iter = 100$) for all PAs (Garrido et al., 2013). According to the first PA method (PA-F), the number of factors that should be retained is equal to the number of *factor* eigenvalues that are greater than the corresponding eigenvalues of appropriately constructed random data (returned when *nfact* is requested). For the second PA method (PA-C), the number of factors that should be retained is equal to the number of *component*⁴ eigenvalues that are greater than

³ Components are linear composites of the original observed variables (i.e., indicators) which can include both common and unique variance (see Byrne, 2005; Fabrigar et al., 1999; Preacher & MacCallum, 2003; Widaman, 2018). Therefore, components from a PCA are not equivalent to factors from an EFA. However, for convenience, in the remainder of this article, we refer to both factors from EFA and components from PCA as “factors.”

⁴ What is meant by “component” here is the same as “component” in PCA.

the corresponding eigenvalues of appropriately constructed random data (returned when *ncomp* is requested).

The second decision point we simulated was the choice of EFA or PCA for analysis. As previously mentioned, PCA is not EFA nor a type of factor analysis. However, we included it as an option in the simulation because it is often used in place of EFA (e.g., Byrne, 2005; Fabrigar et al., 1999; Preacher & MacCallum, 2003; Widaman, 2018). EFAs and PCAs were estimated with the same number of factors as the generating model independent of the number of factors suggested by PA-F, PA-C, or K1 (more on this in the next section). To estimate EFAs and PCAs, the *fa* and *principal* functions (*psych* package; Revelle, 2019) were used, respectively. As previously mentioned, the number of factors specified for all analyses was equal to the number of factors from the generating model. For EFAs, ordinary least squares estimation was used (*fm* = “*ols*”).

The third decision point we simulated was the choice to retain the full scale or shorten the scale (i.e., scale version). We compared retaining all indicators to dropping 25% of the indicators. For the shortened scale scenario, an initial EFA or PCA was estimated with all indicators. Then, 25% of the indicators with the weakest estimated factor loadings were dropped, and the EFA or PCA was reestimated with the remaining indicators. Although the magnitude of generating loadings was constant, it was possible to drop indicators with the weakest factor loadings since estimates varied within and across samples.

The fourth decision point we simulated was the choice of rotation. We compared oblique quartimin (*rotate* = “*quartimin*”), oblique promax (*rotate* = “*Promax*”), and orthogonal varimax (*rotate* = “*varimax*”) rotations. Because all generating factor correlations were nonzero, a reviewer posed a constructive question regarding whether orthogonal varimax rotation was given a fair chance against the oblique alternatives (e.g., quartimin, promax). To address this question, we conducted a follow-up simulation that included uncorrelated generating factors. The results of this simulation demonstrate that, for the conditions tested, there were no drawbacks to choosing an oblique rotation over an orthogonal rotation because if factors were truly orthogonal, then estimates of factor correlations with an oblique rotation were near zero. This is consistent with the established literature (e.g., Fabrigar et al., 1999; Floyd & Widaman, 1995; Harman, 1976), which guided our choice to exclude conditions with orthogonal generating factors in the primary simulation. The method and results for the follow-up rotation simulation are available in the [online supplemental materials](#).

Ideal Scenarios

It is important to note that aspects of the simulation represent a more ideal case than is likely to be encountered in practice. First, we specified the correct number of factors for all EFAs and PCAs regardless of the number of factors suggested by PA-F, PA-C, and K1. In practice, the consequences of relying on this number of factors method and estimating a model with the “wrong” number of factors would only exacerbate the consequences of subsequent methodological choices. However, our aim was to separate the process for determining the number of factors from subsequent analytical decisions to more clearly visualize the effects of specific decision points. Second, we generated data using a model with independent clustering (each indicator loads onto only one factor). In

practice, the factor loading matrix is likely to be more complex (e.g., cross-loadings). Third, we generated continuous data. In practice, indicators are likely to be categorical in nature (e.g., ordinal rating scales). However, if corrective methods (e.g., polychoric correlations) work well (e.g., Flora & Curran, 2004), then generating continuous data would be a reasonable approximation to categorical data that are appropriately modeled.

Despite these artificially simplified aspects of our simulation, we see this design as a reasonable starting point. In other words, it provides a baseline for how methodological choices can impact EFA results, even in a simulated environment that is less complex than the real world. If issues are identified in these idealized circumstances, it seems reasonable to assume that such issues will be exacerbated in more complex circumstances. The current simulation should therefore be viewed as providing a “conservative” sense of what to expect in practice.

Assessing Recovery

We focused on two aspects of recovery: (a) recovery of the number of factors and (b) recovery of the factor structure.

Number of Factors

First, we assessed the accuracy of PA-F, PA-C, and K1 when using each method to determine the number of factors across the 840,000 generated data sets. A choice between options that lead to different answers will push researchers down different paths and toward different results. As we described in the Methodological Flexibility section, differences in performance *between* methods and *within* methods can increase the risk of injecting avoidable heterogeneity into replications of an EFA. For the number of factors specifically, then, the potential for avoidable heterogeneity is not only dependent on whether the options suggest a plausible number of factors but also on the range of number of factors that the selected option suggests across replications.

Factor Structure

Second, we evaluated three recovery measures for factor structure. The first measure was factor loading congruence, which we assessed using the following congruence coefficient (*K*; MacCallum et al., 1999, 2001):

$$r_k = \frac{\sum_{j=1}^p \lambda_{jk(s)} \lambda_{jk(g)}}{\sqrt{\left(\sum_{j=1}^p \lambda_{jk(s)}^2\right) \left(\sum_{j=1}^p \lambda_{jk(g)}^2\right)}} \quad (2)$$

$$K = \frac{\sum_{k=1}^m r_k}{m} \quad (3)$$

As shown in Equation 2, a measure of congruence (*r*) was computed for each factor (*k*) between estimated loadings from the sample ($\lambda_{jk(s)}$) and generating loadings ($\lambda_{jk(g)}$). As shown in Equation 3, *r_k* was then averaged across the number of factors (*m*) to compute the summary measure *K* (i.e., factor loading congruence). Essentially, *K* is the correlation between estimated and generating loadings averaged across factors.

MacCallum et al. (1999, 2001) rotated the sample solution using the generating structure as a target matrix and used the sample loadings from the target rotation to compute K (we refer to this as K_G). Additionally, MacCallum et al. (2001) used the sample loadings from an oblique quartimin rotation to compute K and found the same general pattern of results as for K_G . For the current study, we computed K with solutions from target rotations (K_G) and oblique quartimin rotations as was done in MacCallum et al. (2001). We extended this approach to include oblique promax rotation and orthogonal varimax rotation. In total, we rotated sample solutions in four different ways: (a) oblique quartimin, (b) oblique promax, (c) orthogonal varimax, and (d) oblique target. The target rotation (K_G) provides a benchmark for statistical performance because the generating structure (correct answer) is known and the sample solution is rotated toward it. This optimizes congruence between generating and sample solutions. K_G was used as a baseline measure of factor structure recovery and compared to all other rotations using ΔK , which was computed as the difference between K_G and K for each rotation. Thus, K_G represents the closest achievable solution when the generating model is known and ΔK represents departures from this for the “blind,” or nontarget rotations.

According to MacCallum et al. (1999, 2001), higher values of K are indicative of better factor structure recovery. They provided the following guidelines for the interpretation of K which were used to evaluate factor loading congruence for the current simulation: *Excellent* = .98–1.00, *Good* = .92–.98, *Borderline* = .82–.92, *Poor* = .68–.82, and *Terrible* < .68. These categories are not meant to serve as strict cutoffs but are used as a guide to help facilitate the interpretation of results.

The second and third measures used to evaluate factor structure recovery were mean bias and root mean square error (RMSE) of factor correlations. The mean bias was computed as the average bias (estimated factor correlation minus corresponding generating factor correlation) across factors for each simulation replication. Mean bias and RMSE values closer to zero are indicative of better factor correlation recovery. As with the number of factors methods, the potential for avoidable heterogeneity increases as the recovery of factor loadings or factor correlations worsens, particularly when researchers vary in whether their selected option provides good recovery.

Effect Sizes for Researcher Degrees of Freedom and Questionable Research Practices

We used ANOVA models to assess the impact of manipulated data features (sample size, the number of factors, the number of indicators per factor, the magnitude of factor correlations, and the magnitude of factor loadings) and methodological choices (number of factors method, scale version, analysis, and rotation) on the accuracy of determining the correct number of factors, factor loading congruence, and factor correlation recovery. Main effects and interactions up to three-way interactions were included in the analyses. Effect sizes (partial η^2 s) are presented for number of factors methods in Table 1, factor loading congruence in Table 2, and factor correlation recovery in Table 3. To ensure that the most influential effects were of focus, we probed all partial η^2 s greater than or equal to 0.06 (medium effect; Cohen, 1988). All other effects were omitted in Tables 1–3.

Table 1

Effect Sizes: Main Effects and Interactions for Number of Factors Methods

Effect	Partial η^2_{PA-F}	Partial η^2_{PA-C}	Partial η^2_{K1}
Sample size (N)	0.87	0.82	0.96
Number of factors (#)	—	0.06	0.75
Factor correlation (φ)	0.34	0.57	—
Factor loading (λ)	0.37	0.43	0.97
Indicators per factor (p)	0.30	0.34	0.63
# $\times$ p	—	—	0.39
# $\times$ φ	0.12	0.21	—
$\lambda \times$ #	0.22	0.21	0.87
$\lambda \times p$	—	—	0.72
$\lambda \times \varphi$	0.23	0.20	—
$N \times$ #	0.17	0.18	0.19
$N \times p$	—	—	0.11
$N \times \varphi$	0.16	0.15	—
$\lambda \times N$	0.19	—	0.54
$\lambda \times p \times \#$	—	—	0.30
$\lambda \times N \times \#$	0.07	—	0.33
$\lambda \times N \times p$	—	—	0.18
$\lambda \times N \times \varphi$	0.14	0.20	—

Note. Partial η^2_{PA-F} = effect size on the suggested number of factors when using parallel analysis with factor eigenvalues; Partial η^2_{PA-C} = effect size on the suggested number of factors when using parallel analysis with component eigenvalues; Partial η^2_{K1} = effect size on the suggested number of factors when using the Kaiser–Guttman criterion. Effect sizes < .06 (medium; Cohen, 1988) were omitted.

Results

Given the large volume of results, the following paragraph describes our general organization of the results section. We begin by describing results for the number of factors. Specifically, we look at how well PA-F, PA-C, and K1 recovered the generating

Table 2

Effect Sizes: Main Effects and Interactions for Factor Loading Congruence (K and ΔK)

Effect	Partial η^2_K	Partial $\eta^2_{\Delta K}$
Sample size (N)	0.26	—
Number of factors (#)	0.38	0.06
Factor correlation (φ)	0.33	0.06
Factor loading (λ)	0.57	0.13
Scale version	—	0.11
Rotation	0.12	0.26
$N \times$ #	0.07	—
$\varphi \times$ Rotation	—	0.10
$\lambda \times$ Version	—	0.11
$\lambda \times$ Rotation	—	0.11
$\lambda \times N$	0.26	—
$\lambda \times \#$	0.23	—
$\lambda \times \varphi$	0.13	—
$\lambda \times N \times \#$	0.06	—

Note. Partial η^2_K = effect size on the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using oblique quartimin, oblique promax, and orthogonal varimax (K); Partial $\eta^2_{\Delta K}$ = effect size on the difference between K and K_G , the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using the generating solution as a target rotation (optimal congruence). Effect sizes < .06 (medium; Cohen, 1988) were omitted.

Table 3

Effect Sizes: Main Effects and Interactions for Factor Correlation Bias and RMSE

Effect	Partial η^2_{Bias}	Partial η^2_{RMSE}
Sample size (N)	0.28	0.38
Number of factors (#)	0.10	0.11
Factor correlation (φ)	0.85	0.85
Factor loading (λ)	0.72	0.73
Analysis	0.12	0.09
Rotation	0.95	0.96
Rotation $\times$ Analysis	0.07	—
Rotation $\times N$	0.17	0.24
Rotation $\times \#$	—	0.07
Rotation $\times \varphi$	0.65	0.73
Rotation $\times \lambda$	0.59	0.60
$N \times \lambda$	0.07	—
$\varphi \times \lambda$	0.26	0.37
Rotation $\times \varphi \times \lambda$	0.16	0.24

Note. Partial η^2_{Bias} = effect size on the mean bias of estimated factor correlations; Partial η^2_{RMSE} = effect size on root mean square error of estimated factor correlations. Effect sizes $< .06$ (medium; Cohen, 1988) were omitted.

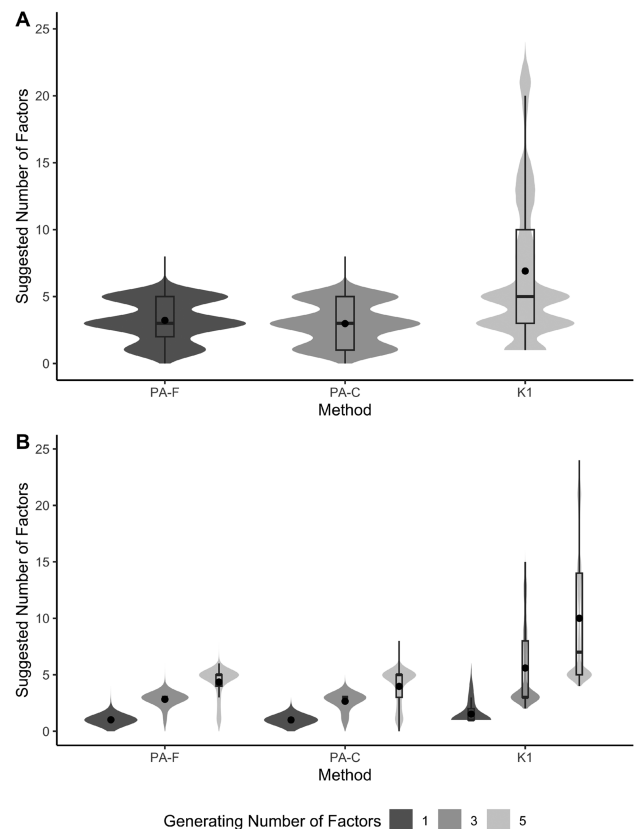
number of factors (may contribute to between-decision heterogeneity) as well as the range of number of factors suggested by each method (a form of within-decision heterogeneity). We then move on to recovery measures for factor structure. In this section, we present the three-way interactions, two-way interactions, and main effects that were influential for factor loading congruence. We also provide frequencies for the different levels of factor loading congruence (*Excellent*, *Good*, *Borderline*, *Poor*, and *Terrible*) based on the guidelines for interpreting K in MacCallum et al. (1999, 2001). To conclude the results, we present three-way interactions, two-way interactions, and main effects that were influential for factor correlation recovery. Following the focal simulation study results, we focus more narrowly on an example simulation replication to pedagogically demonstrate the impact of methodological choices on the heterogeneity of EFA results.

Number of Factors

With the exception of the generating number of factors on PA-F and the magnitude of factor correlations on K1, all manipulated data features were influential on the number of factors suggested by PA-F, PA-C, and K1 (Table 1). The accuracy of each method is summarized in Figure 1. Panel A shows the number of factors suggested by PA-F, PA-C, and K1 aggregated across all manipulated data features (sample size, the number of factors, the magnitude of factor correlations, the magnitude of factor loadings, and the number of indicators per factor). Given that distributions in panel A peaked around the generating number of factors (one, three, and five), panel B shows the suggested number of factors for each method by the generating number of factors. Accuracy rates averaged over replications for each condition are available in Table S1 in the online supplemental materials, and similar to panel B of Figure 1, panels A–D of Figure S1 in the online supplemental materials display the suggested number of factors for PA-F, PA-C, and K1 by sample size, the magnitude of factor correlations, the magnitude of factor loadings, and the number of indicators per factor, respectively.

Figure 1

Suggested Number of Factors for PA-F, PA-C, and K1



Note. PA-F = parallel analysis with factor eigenvalues; PA-C = parallel analysis with component eigenvalues; K1 = Kaiser–Guttman criterion. The dots within the boxplots represent the mean for each condition. Because distributions in panel A peaked around the generating number of factors (one, three, and five), panel B plots the suggested number of factors (y-axis) for each method by the generating number of factors (grouped by color).

All three number of factors methods had greater accuracy for larger sample sizes and fewer factors. The PA methods performed better with more indicators, smaller factor correlations, and larger factor loadings. The agreement was high between PA-F and PA-C (approximately 84%), but PA-F was more accurate across conditions overall. For K1, accuracy increased with fewer indicators. Although K1 was consistent across the varying levels of factor correlations, it was highly inconsistent across the varying levels of factor loadings. It achieved nearly perfect recovery when loadings were strong (0.89) but was nearly always incorrect when loadings were weak (0.45). When comparing the accuracy between PA methods and K1, both PA methods consistently outperformed K1. Out of the 840,000 unique data sets generated, PA-F suggested the correct number of factors for 82% of data sets. When incorrect, PA-F under-factored 94% of the time. PA-C suggested the correct number of factors for 73% of data sets. When incorrect, PA-C under-factored more than 99% of the time. K1 suggested the correct number of factors for 53% of data sets. When incorrect, K1 over-factored nearly 100% of the time.

The ranges of number of factors suggested across methods as well as the mean number of factors over or under the generating number are summarized in Table 4.⁵ The results comparing PA-F with PA-C were more similar than either PA method compared to K1, and both PA methods had a relatively narrow range of recommended answers (illustrated in Figure 1). For five factors, the number of factors most difficult to recover correctly, PA-F and PA-C only over-factored by an average of one and under-factored by an average of no more than three. On the other hand, the range of answers for K1 was quite wide (also illustrated in Figure 1). Although K1 only under-factored by an average of one, it over-factored by as much of an average of nine. The wide range of number of factors suggested by K1 exemplifies that, in addition to heterogeneity from making different choices among K1 and PA methods (between-decision heterogeneity), additional heterogeneity is measurable even among researchers who all select K1 (within-decision heterogeneity).

Factor Loading Congruence (K and ΔK)

When simulating the practice of dropping 25% of the indicators for conditions in which the scale was shortened, there was no limit set on how many indicators could be dropped from each factor. As a result, there were rare instances (0.003% of replications with five factors for the generating model and a shortened scale) where the remaining indicators for a factor all had corresponding generating values equal to zero. Prior to computing K , this generating factor loading vector as well as the corresponding loading vector in the sample were removed, and K was computed with the remaining factors. This adjustment was made to avoid a denominator of zero for r_k in Equation 2 due to the generating sum of squared loadings being equal to zero. Although infrequent, it still is worth noting the possibility of this happening in practice. When shortening multidimensional scales, blindly dropping indicators using arbitrary rules (e.g., factor loading cutoffs, target percentages) could lead to issues such as these and be misleading in terms of the number of factors (in this case, under-factoring).

Table 4

Summary of Range and Over-/Under-Factoring for PA-F, PA-C, and K1

Method	Number of factors	Minimum suggestion	Maximum suggestion	Mean over-factor	Mean under-factor
PA-F	1	0	4	1.06	-1.00
PA-C	1	0	3	1.01	-1.00
K1	1	1	6	1.87	NA
PA-F	3	0	6	1.06	-1.46
PA-C	3	0	5	1.04	-1.52
K1	3	2	16	5.55	-1.00
PA-F	5	0	8	1.06	-2.45
PA-C	5	0	8	1.06	-2.67
K1	5	4	24	9.21	-1.00

Note. PA-F = parallel analysis with factor eigenvalues; PA-C = parallel analysis with component eigenvalues; K1 = Kaiser-Guttman criterion; NA = did not under-factor. Although some minimum suggestions for the number of factors were zero, a researcher would not estimate a model with zero factors. When considering the range of number of factors, practically, the minimum suggestion should be one factor.

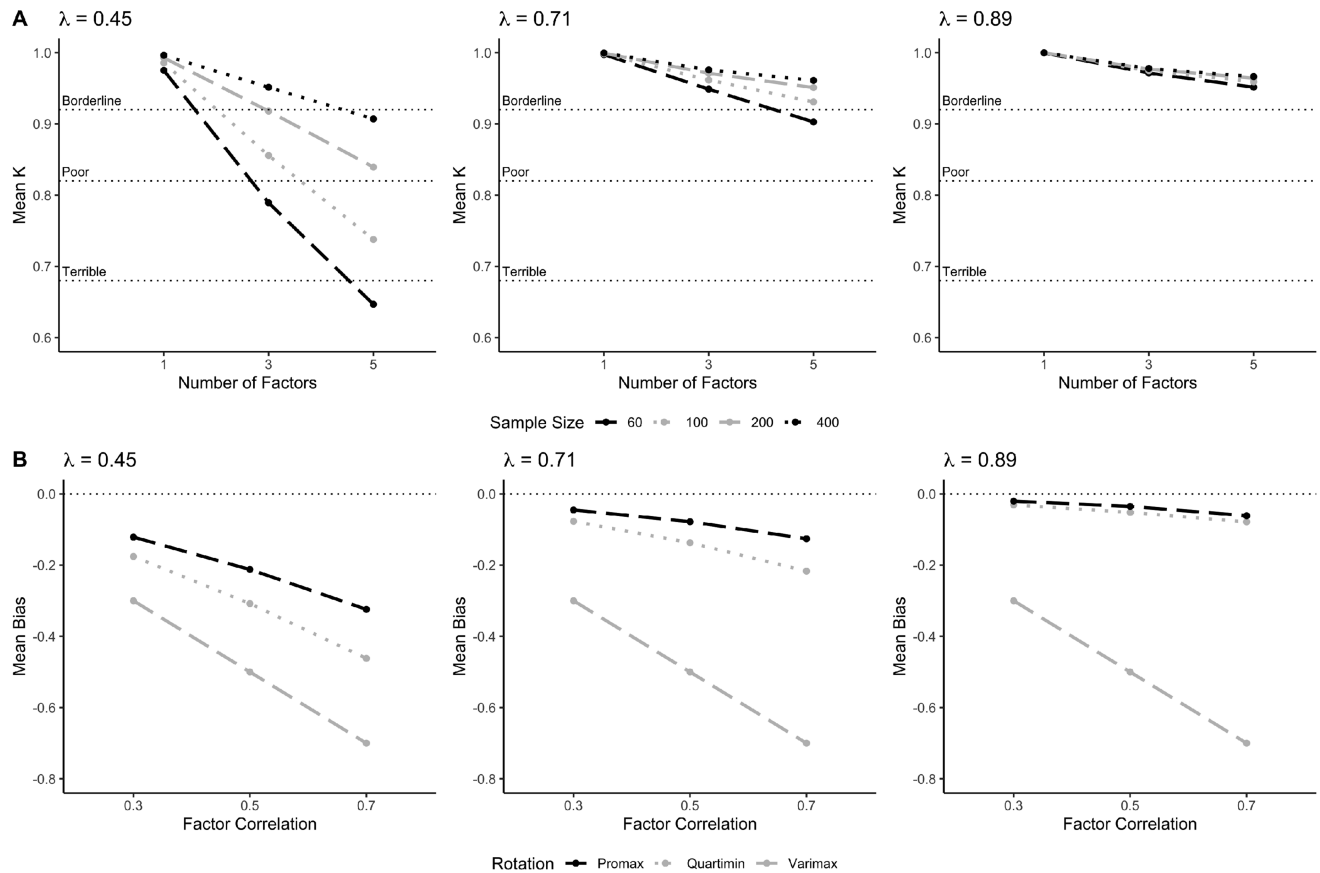
Effects on Factor Loading Congruence

First, we examine the three-way interaction for K , the congruence coefficient computed with loadings from oblique quartimin, oblique promax, and orthogonal varimax rotations, with an effect size greater than or equal to 0.06 (according to Table 2). Panel A of Figure 2 plots the interaction between the number of factors, sample size, and the magnitude of factor loadings. Recovery was mostly acceptable ($K \geq 0.92$) across sample size and the number of factors when generating factor loadings were 0.89 or 0.71. However, when generating factor loadings were 0.45, congruence for almost all three- and five-factor conditions fell below 0.92. One-factor models were relatively unaffected across the varying magnitudes of factor loadings and sample size. There were no three-way interactions for ΔK , the distance from optimal congruence, that were greater than or equal to 0.06.

Second, we examine two-way interactions for K and ΔK . The seven two-way interactions for K and ΔK with effect sizes greater than or equal to 0.06 are displayed in Figure 3. Panels A–D plot two-way interactions for K . Panels A–C of Figure 3 echo the results for the three-way interaction in panel A of Figure 2. Congruence almost always fell below 0.92 for conditions with more than one factor or weak factor loadings (0.45). In panel D of Figure 3, congruence fell below 0.92 when factor loadings were weak (0.45) regardless of the magnitude of factor correlations. Panels E–G of Figure 3 plot two-way interactions for ΔK . Panels E–F display interactions that include rotation. Oblique promax and oblique quartimin rotations performed relatively well, and performance was similar between the two methods. In contrast, orthogonal varimax performed the worst overall and diverged from optimal congruence as the magnitude of factor correlations and factor loadings increased. This was expected because orthogonal rotations constrain factor correlations to zero. Panel G plots the interaction between the magnitude of factor loadings and scale version. The only noticeable difference between the full and short scale was when factor loadings were weak (0.45).

Third, we examine the main effects for K and ΔK (Figure 4). Panels A–E of Figure 4 display the main effects for K with effect sizes greater than or equal to 0.06. Congruence decreased as the number of factors and the magnitude of factor correlations increased. Congruence also decreased as the magnitude of factor loadings and sample size decreased. In terms of methodological choices, the use of orthogonal varimax rotation, as opposed to oblique quartimin or oblique promax, led to worse congruence. Panels F–J of Figure 4 display the main effects for ΔK with effect sizes greater than or equal to 0.06. Panels F (number of factors), G (the magnitude of factor correlations), and J (rotation) mirrored the pattern of results for K . However, unique to ΔK , the trend reversed for the magnitude of factor loadings (ΔK increased as magnitude increased), and the effect size for scale version was greater than 0.06. Specifically, the choice of dropping indicators as opposed to retaining the full scale resulted in greater congruence. However, we note that recovery was better in this case because estimation was artificially made simpler. This result was obtained not simply because indicators were dropped

⁵ Although some minimum suggestions for the number of factors reported in Table 4 were zero, a researcher would not estimate a model with zero factors. When considering the range of number of factors, practically, the minimum suggestion should be one factor.

Figure 2*Three-Way Interactions for Factor Loading Congruence (K) and Factor Correlation Bias*

Note. K = the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using oblique quartimin, oblique promax, and orthogonal varimax; Bias = estimated factor correlation minus generating factor correlation. Panel A displays the interaction between the number of factors and sample size across the different magnitudes for factor loadings (λ). The y-axis is mean K , and the thresholds for *Borderline*, *Poor*, and *Terrible* are based on the MacCallum et al. (1999, 2001) guidelines for interpreting congruence. Panel B displays the interaction between the magnitude for factor correlations and rotation across the different magnitudes for factor loadings (λ). The y-axis is mean bias, and the horizontal dotted line is plotted at zero (no bias). Error bars were omitted because all standard errors were $\leq .001$. Because the orthogonal varimax rotation constrained factor correlations to zero, the points plotted for varimax as a separate line (panel B) were equal to the magnitude of the average generating factor correlation in each condition.

but because of *how* indicators were dropped (further detailed in the Discussion section).

Falling Below “Good” Recovery

Across all replications, approximately 42% of replications achieved *Excellent* recovery, 28% achieved *Good* recovery, 19% had *Borderline* recovery, 6% had *Poor* recovery, and 5% had *Terrible* recovery, according to the guidelines for interpreting K in MacCallum et al. (1999, 2001). Frequencies of recovery levels for each simulation variable (data feature or decision point) with an effect size on K or ΔK greater than or equal to 0.06 are displayed in Figure 5. There is one panel for each such simulation variable and each panel shows, for every level within the simulation variable, the frequency of *Excellent/Good/Borderline/Poor/Terrible* recovery.

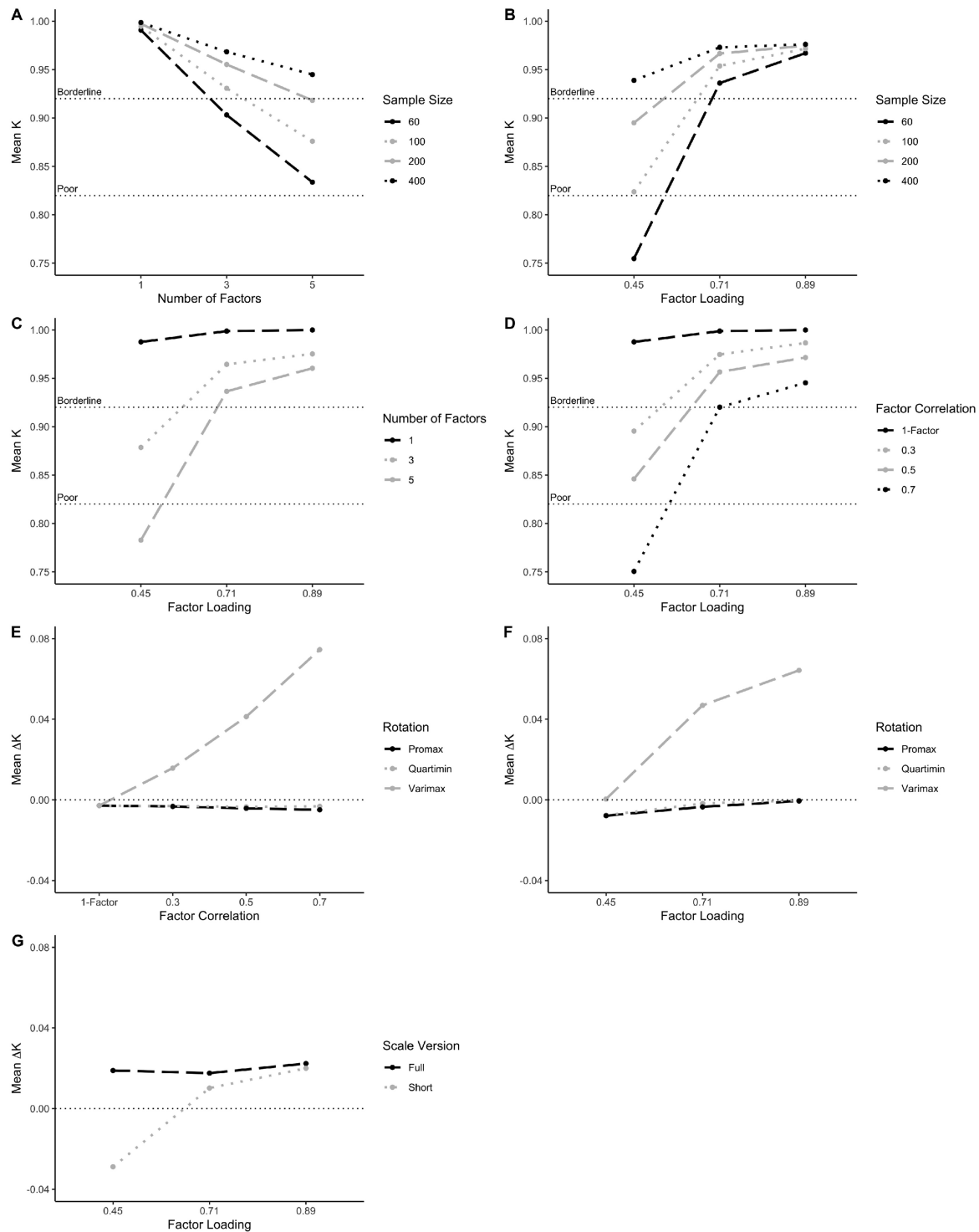
Figure 5 helps visually distinguish between conditions with more or less variability in congruence. For example, the levels within factor loading (panel D), were quite different from one another with the

strongest factor loadings resulting in the greatest proportion of *Excellent* recovery. In contrast, there was almost no difference across levels for scale version (panel E). Additionally, Figure 5 helps visually distinguish variability between levels within a condition. For example, in panel F, the distributions for oblique quartimin and oblique promax were nearly identical. However, the distribution for orthogonal varimax was perceptibly different. The frequency of *Excellent* recovery decreased while the frequency of *Borderline* recovery increased.

Factor Correlation Recovery

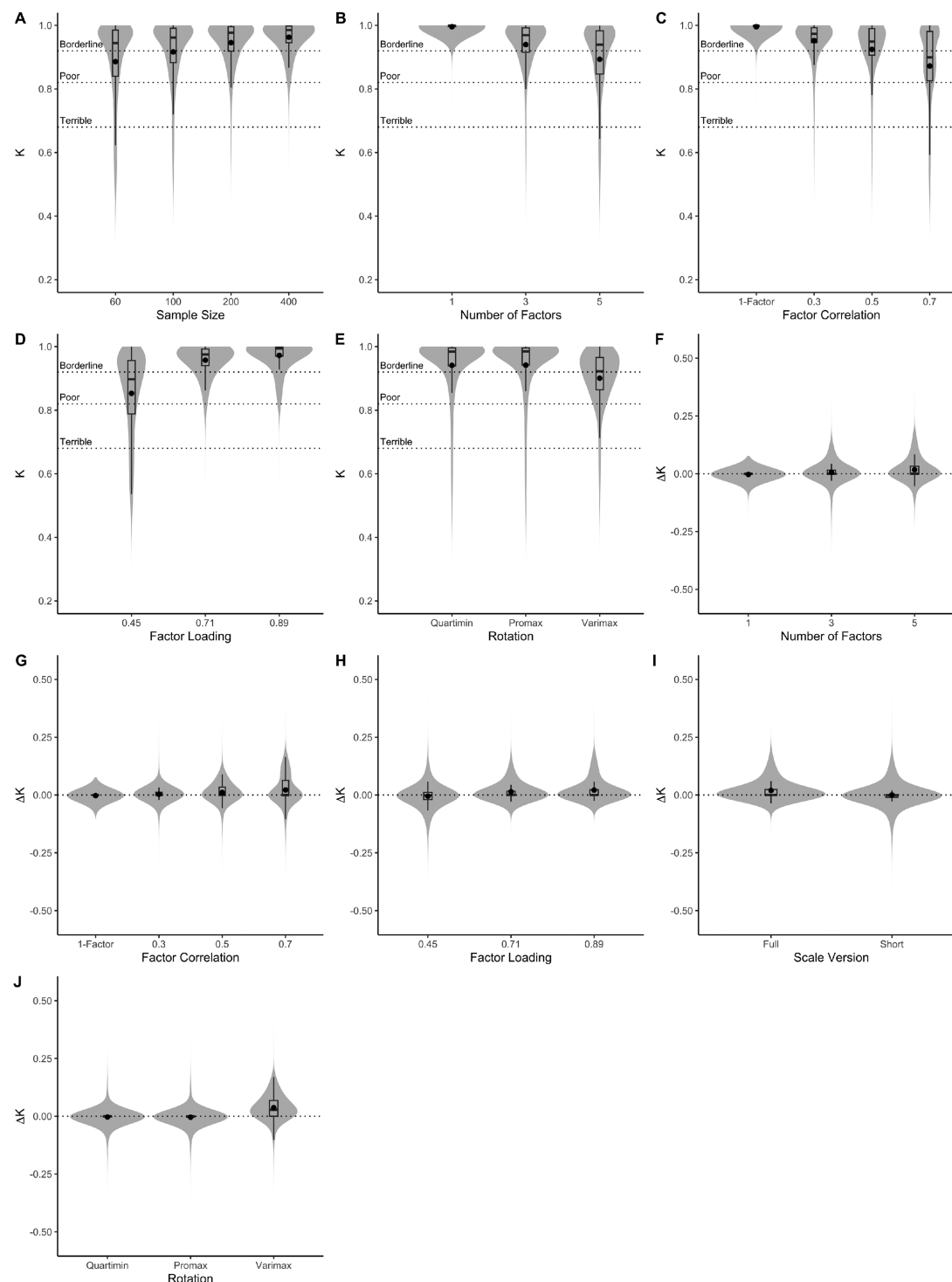
There was one three-way interaction for factor correlation bias with an effect size greater than 0.06. Panel B of Figure 2 plots the interaction between the magnitude of factor correlations, rotation, and the magnitude of factor loadings. Bias increased for oblique quartimin and oblique promax rotations as the magnitude of factor correlations increased and the magnitude of factor loadings

Figure 3
Two-Way Interactions for Factor Loading Congruence (K and ΔK)

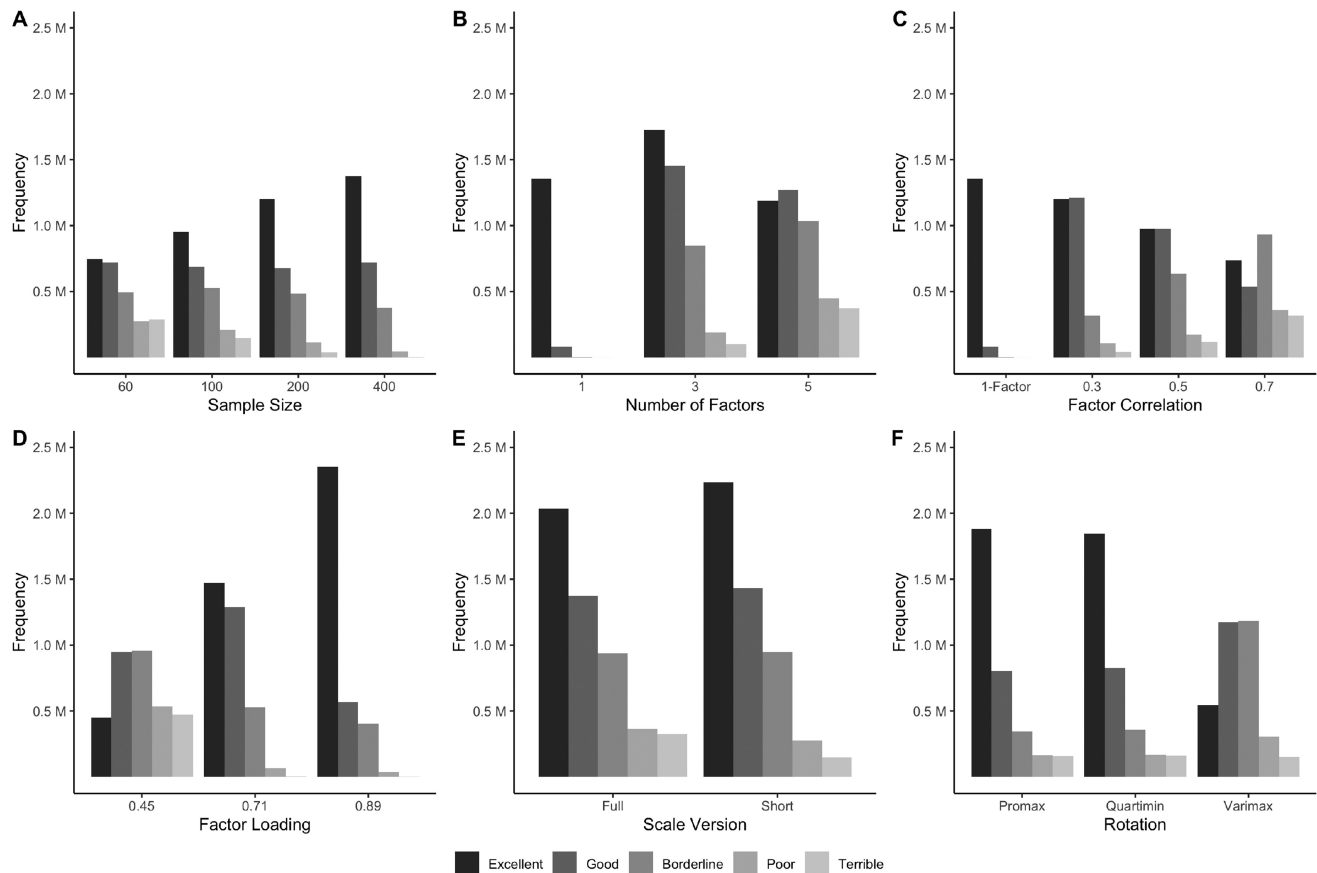


Note. K = the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using oblique quartimin, oblique promax, and orthogonal varimax; ΔK = the difference between K and K_G , the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using the generating solution as a target rotation (optimal congruence). The y-axes for panels A–D are mean K , and the thresholds for *Borderline* and *Poor* are based on the [MacCallum et al. \(1999, 2001\)](#) guidelines for interpreting congruence. The y-axes for panels E–G are ΔK , and the horizontal dotted line is plotted at zero (at optimal congruence). Error bars were omitted because all standard errors were $\leq .001$.

Figure 4
Main Effects for Factor Loading Congruence (K and ΔK)



Note. K = the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using oblique quartimin, oblique promax, and orthogonal varimax; ΔK = the difference between K and K_G , the congruence coefficient computed between generating factor loadings and estimated factor loadings rotated using the generating solution as a target rotation (optimal congruence). The dots within the boxplots represent the mean for each condition. The y-axes for panels A–E are K , and the thresholds for *Borderline*, *Poor*, and *Terrible* are based on the MacCallum et al. (1999, 2001) guidelines for interpreting congruence. The y-axes for panels F–J are ΔK , and the horizontal dotted line is plotted at zero (at optimal congruence).

Figure 5*Frequency of Excellent, Good, Borderline, Poor, and Terrible Factor Loading Congruence*

Note. Classification as *Excellent*, *Good*, *Borderline*, *Poor*, and *Terrible* is based on the MacCallum et al. (1999, 2001) guidelines for interpreting congruence.

decreased. The performance also diverged between the two oblique rotations as the magnitude of factor loadings decreased. Estimated factor correlations were consistently most biased when using an orthogonal varimax rotation. Again, this was expected because orthogonal rotations constrain factor correlations to zero. This can be seen in panel B of Figure 2 where the mean bias for orthogonal varimax rotation, regardless of the magnitude of factor loadings, was equal to the magnitude of generating factor correlations.

Two-way interactions for factor correlation bias with an effect size greater than or equal to 0.06 are displayed in Figure 6. In panels A–D, rotations are plotted as separate lines. Oblique quartimin and oblique promax performed similarly across conditions. For the two oblique rotations, recovery worsened as sample size and the magnitude of factor loadings decreased, as the magnitude of factor correlations increased, and when PCA was used instead of EFA. Again, and as expected, orthogonal varimax consistently performed worse compared to the oblique alternatives. Similar to the three-way interaction in panel B of Figure 2, bias for orthogonal varimax was equal to the magnitude of the average generating factor correlation for each condition. Panels E and F of Figure 6 show that the effect of sample size and the magnitude of factor correlations on bias, respectively, was greater as the magnitude of factor loadings decreased.

Figure 7 displays the main effects of factor correlation bias with effect sizes greater than or equal to 0.06. For data features, the pattern of results

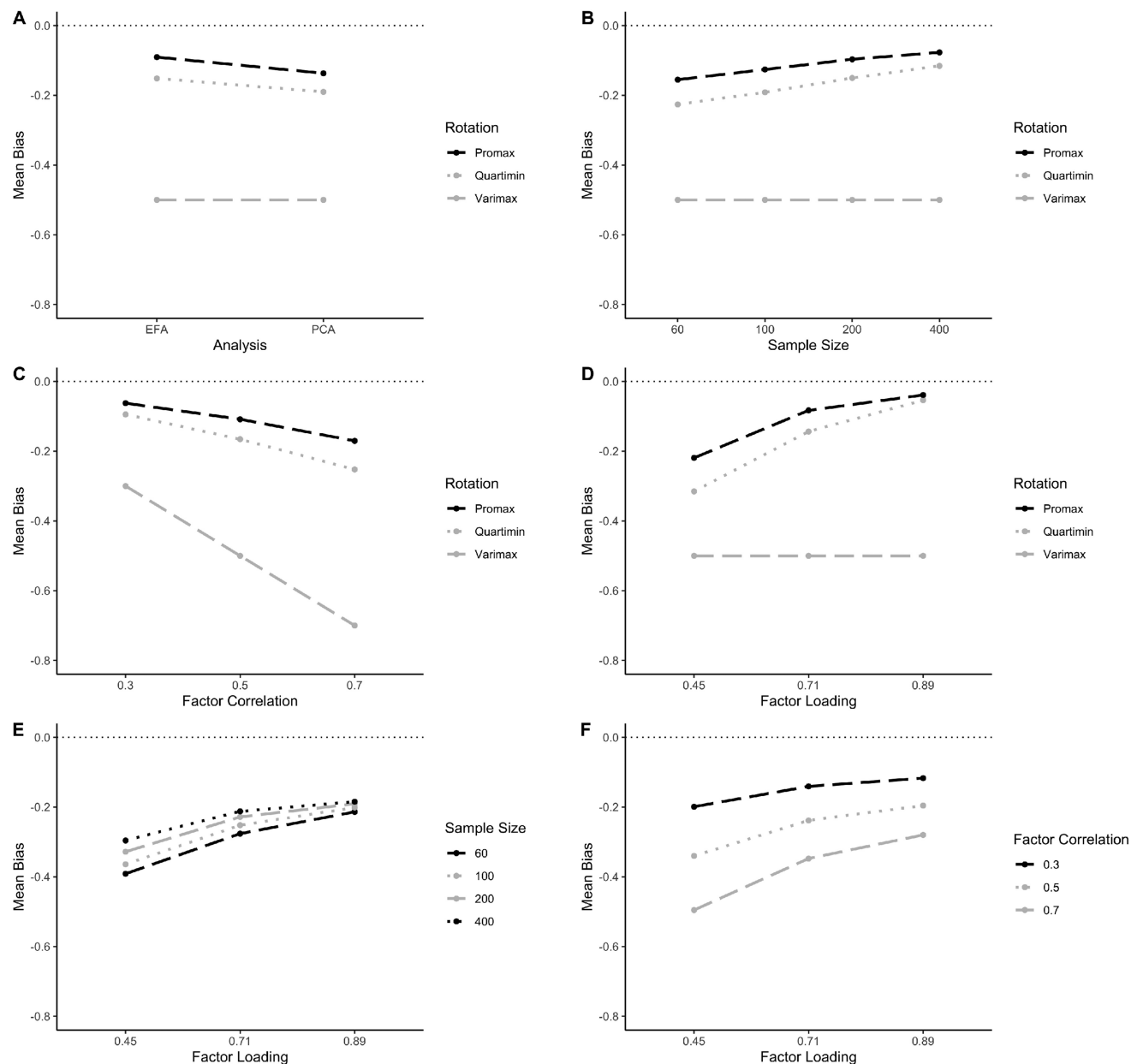
was the same as for factor loading congruence. Recovery worsened as the number of factors and the magnitude of factor correlations increased. Recovery also worsened as the magnitude of factor loadings and sample size decreased. For methodological choices, there was greater bias when using PCA (as opposed to EFA). Among choices for rotation, oblique promax resulted in the least bias, oblique quartimin resulted in slightly more bias, and orthogonal varimax resulted in the most bias.

Finally, the three-way interaction, two-way interactions, and main effects for factor correlation RMSE with effect sizes greater than or equal to 0.06 are available in Figures S2–S4, respectively, in the online supplemental materials. Differences in recovery were generally driven by bias, so RMSE did not offer any new recovery patterns. There was only one unique two-way interaction for RMSE with an effect size greater than 0.06 (the number of factors $\times$ rotation). Otherwise, if an interaction or main effect for RMSE had an effect size greater than or equal to 0.06, there was a corresponding interaction or main effect for bias with an effect size greater than or equal to 0.06 and a pattern of results that matched exactly.

An Example of Heterogeneity in EFA Results

To demonstrate the impact of methodological choices on the heterogeneity of EFA results, we pull one replication from the simulation detailed above (simulated data can be accessed at

Figure 6
Two-Way Interactions for Factor Correlation Bias



Note. Bias = estimated factor correlation minus generating factor correlation. The horizontal dotted line is plotted at zero (no bias). Error bars omitted because all standard errors were $\leq .001$. Because the orthogonal varimax rotation constrained factor correlations to zero, the points plotted for varimax as a separate line (panels A–D) were equal to the magnitude of the average generating factor correlation in each condition.

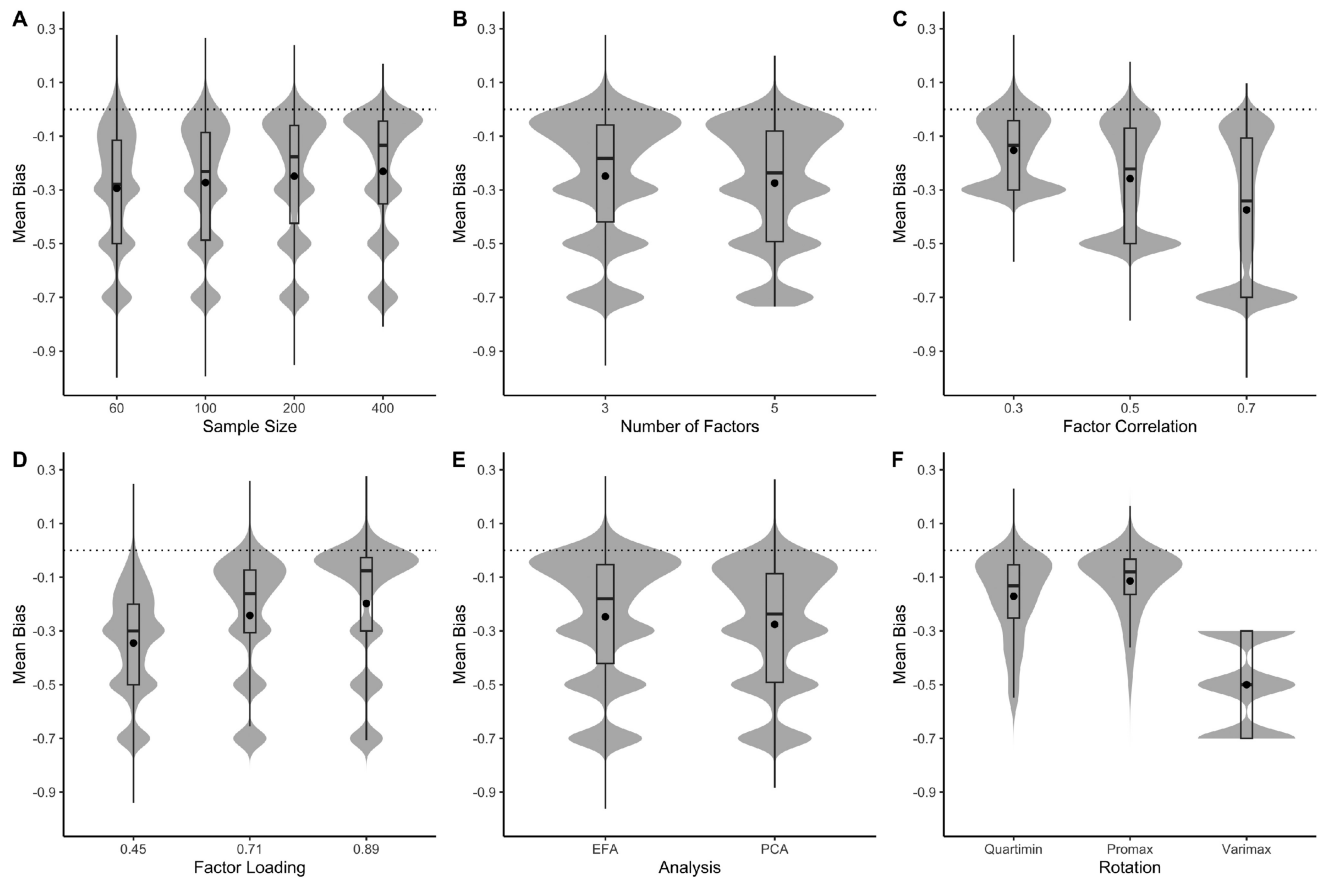
<https://github.com/patrickmanapat/efaHet>) and more closely examine the results from two analyses with different options.

The replication we used was drawn from the following design cell: $N = 100$, three factors, eight indicators per factor, $\varphi = 0.3$, and $\lambda = 0.71$. The decision points for the analyses were: (a) method for determining the number of factors, (b) EFA or PCA, and (c) rotation. We refer to the first set of decisions as Path 1, which included (a) PA-F, (b) EFA, and (c) oblique promax rotation. These practices performed well in the current simulation. Alternatively, we refer to the second

set of decisions as Path 2, which included (a) K1, (b) PCA, and (c) orthogonal varimax rotation. For at least one measure of recovery, these practices performed poorly compared to other options. This set of practices is also known as the “Little Jiffy” (Kaiser, 1970) and despite being recommended against in the psychometric literature (see, e.g., Preacher & MacCallum, 2003), remains widely used in practice and the default in some software.

The first decision point was the choice of method to determine the number of factors. PA-F for Path 1 suggested three factors, the correct

Figure 7
Main Effects for Factor Correlation Bias



Note. Bias = estimated factor correlation minus generating factor correlation. The dots within the boxplots represent the mean for each condition. The horizontal dotted line is plotted at zero (no bias). Constraining factor correlations to zero with the orthogonal varimax rotation led to the peaks in distributions around the magnitudes of generating factor correlations (0.3, 0.5, and 0.7; see panel F).

number of factors. K1 for Path 2, on the other hand, suggested four factors (over-factored). Even at this initial decision point, for the most basic aspect of the factor model (the number of factors), different choices suggested different answers. For the second decision point, we retreat back to the correct number of factors as is done in the simulation to keep the decision for number of factors methods and subsequent decisions regarding the EFA or PCA separate. An EFA with three factors was estimated for Path 1, and the solution was rotated using oblique promax. For Path 2, a PCA with three factors was estimated, and the solution was rotated using orthogonal varimax.

The estimated factor loadings and factor correlations for Path 1 and Path 2, as well as the generating factor structure, are provided in Table 5. To ease interpretation, all loadings with a magnitude greater than or equal to 0.4 are in bold. Clearly, Path 1 and Path 2 resulted in very different solutions even when the correct number of factors was specified. Path 1 closely resembles the generating structure ($K = 0.99$, *Excellent*) and its structure suggests it would lead to easier interpretation. This would help facilitate consistent interpretations across replications. Path 2, however, did not recover the generating structure well ($K = 0.76$, *Poor*) and would likely prove more difficult to interpret. This simple example shows how methodological flexibility can lead to heterogeneity across results,

even when conducting analyses on the *same dataset*, implicitly controlling for sampling variability and other sources of heterogeneity.

Discussion

The main purpose of the current study was to assess the impact of methodological flexibility on EFA results. Psychometrics is a critical part of psychological science, and the validity of research findings hinges, at least in part, on the quality of the measures used (Flake et al., 2017, 2022; Flake & Fried, 2020; Hussey & Hughes, 2020; Shaw et al., 2020). The use of a scale is supported if there is reasonable evidence of score reliability and appropriate validity. However, the extent to which we can rely on that evidence depends on how well psychometric results replicate. Yet, psychometrics is often left out of the replication discussion (Flake & Fried, 2020; Pargent et al., 2019; Shaw et al., 2020). In the current study, we sought to highlight psychometrics in the context of replication and examine what contributes to heterogeneity across a series of EFA results. Before discussing EFA decision points with the greatest potential for injecting heterogeneity into results, we summarize the key findings for recovery of the number of factors as well as the factor structure.

Table 5*Example of Heterogeneous Results From Forking Paths in EFA*

Generating factor structure Factor loadings			Path 1 estimates: EFA and promax Factor loadings			Path 2 estimates: PCA and varimax Factor loadings		
F1	F2	F3	F1	F2	F3	F1	F2	F3
0.71	0	0	0.69	0.01	−0.03	0.53	0.18	0.48
0.71	0	0	0.67	0.04	0.03	0.49	0.16	0.52
0.71	0	0	0.70	−0.03	0.03	0.53	0.11	0.51
0.71	0	0	0.66	0.08	0.09	0.44	0.16	0.59
0.71	0	0	0.68	−0.03	−0.11	0.57	0.19	0.38
0.71	0	0	0.60	−0.01	0.14	0.44	0.04	0.54
0.71	0	0	0.72	−0.12	−0.09	0.62	0.12	0.37
0.71	0	0	0.61	0.06	0.06	0.43	0.15	0.53
0	0.71	0	−0.19	0.72	0.17	−0.46	0.44	0.41
0	0.71	0	0.02	0.61	−0.08	−0.21	0.57	0.30
0	0.71	0	0.08	0.69	−0.07	−0.19	0.62	0.39
0	0.71	0	0.08	0.59	−0.05	−0.17	0.55	0.36
0	0.71	0	−0.05	0.71	0.08	−0.33	0.51	0.42
0	0.71	0	0.03	0.70	−0.04	−0.24	0.59	0.39
0	0.71	0	0.10	0.69	−0.03	−0.19	0.60	0.43
0	0.71	0	−0.06	0.72	0.01	−0.33	0.56	0.37
0	0	0.71	0.02	−0.10	0.62	−0.10	−0.47	0.49
0	0	0.71	0.11	0.08	0.70	−0.12	−0.34	0.69
0	0	0.71	−0.06	−0.02	0.74	−0.22	−0.48	0.56
0	0	0.71	−0.05	0.06	0.76	−0.24	−0.42	0.62
0	0	0.71	0.01	0.03	0.73	−0.18	−0.42	0.62
0	0	0.71	−0.02	−0.02	0.71	−0.18	−0.46	0.56
0	0	0.71	0.00	−0.12	0.72	−0.13	−0.54	0.53
0	0	0.71	0.07	0.06	0.76	−0.15	−0.40	0.70
Factor correlations			Factor correlations			Factor correlations		
F1	F2	F3	F1	F2	F3	F1	F2	F3
1			1			1		
0.30	1		0.29	1		0	1	
0.30	0.30	1	0.27	0.06	1	0	0	1

Note. EFA = exploratory factor analysis; PCA = principal components analysis; F1 = factor 1; F2 = factor 2; F3 = factor 3. Promax was an oblique rotation. Varimax was an orthogonal rotation. Factor loadings with a magnitude greater than or equal to 0.40 are in bold. Based on the MacCallum et al. (1999, 2001) guidelines for interpreting the congruence coefficient (K), factor loading congruence was *Excellent* ($K = 0.99$) for Path 1 and *Poor* ($K = 0.76$) for Path 2 (“Little Jiffy”; Kaiser, 1970). Although Path 1 and Path 2 solutions were estimated specifying the same number of factors/components (three), there was heterogeneity in results for the decision point that preceded the estimation of a factor solution—method for determining the number of factors. Path 1 used parallel analysis with factor eigenvalues which suggested three factors (the correct number), and Path 2 used the Kaiser–Guttman criterion which suggested four factors (incorrect; over-factored).

Summary of Recovery

In terms of data features, recovery was notably worse when factor loadings were weak ($\lambda = 0.45$), there were more factors (five), and the sample size was small ($N = 60$ or $N = 100$). These findings are consistent with previous simulation work (e.g., MacCallum et al., 1999, 2001). In terms of methodological options, method to determine the number of factors and rotation were relatively influential for EFA results. Across methods to determine the number of factors, PA-F was consistently the most accurate, which is consistent with previous findings (e.g., Auerswald & Moshagen, 2019; Lim & Jahng, 2019). Despite the high rate of agreement between PA methods, PA-F slightly outperformed PA-C across conditions. K1, on the other hand, almost always performed the worst out of the three number of factors methods. For rotation, recovery for all options worsened as the magnitude of factor loadings decreased and as the number of factors and magnitude of factor correlations increased. In line with previously established recommendations for rotations in EFA (e.g., Byrne, 2005; Fabrigar et al., 1999; Floyd &

Widaman, 1995; Preacher & MacCallum, 2003), recovery was mostly acceptable (and nearly identical) for oblique quartimin and oblique promax rotations while orthogonal varimax always performed the worst.

The choice between EFA and PCA was also relatively influential but only for recovery of factor correlations, not for factor loading congruence. Although scale version (full vs. shortened) was found to be relatively influential for EFA results, factor loading congruence was greater for the shortened scale conditions because recovery was artificially made simpler. Recovery did not improve just because the scale was shorter. For example, the number of indicators per factor (eight indicators vs. 12 indicators) had effectively no impact on factor loading congruence (partial $\eta^2_K < .01$ and partial $\eta^2_{\Delta K} < .01$). Additionally, as shown in panel G of Figure 3, recovery was roughly the same for the full and shortened scales under the stronger generating factor loadings ($\lambda = 0.71$ and $\lambda = 0.89$). A difference in recovery between scale versions became apparent when the generating factor loadings were weakest. Recovery improved because of *how* scales were shortened.

Specifically, dropping the weakest estimated loadings in conditions where the generating loadings were weakest ($\lambda = 0.45$) led to greater congruence. Since weaker loadings are more difficult to recover (e.g., MacCallum et al., 1999), dropping indicators from conditions with weak generating loadings made recovery easier.

The Potential for Avoidable Heterogeneity

Recovery can also have implications for heterogeneity in results. There is a greater potential for avoidable heterogeneity when researchers select methodological options that differ in performance. In some ways, then, recovery performance can serve as an indicator of methodological options that are at greater risk for contributing heterogeneity. For example, recovery was consistently poor for orthogonal varimax rotation. This finding suggests that there may be utility in eliminating orthogonal varimax rotation as an acceptable option due to both the poor recovery itself and its role in heterogeneity. Although different rotation options would remain, the simulation demonstrated that the choice among the remaining (oblique) alternatives would be less impactful.

Furthermore, this potential for heterogeneity is not limited to a single decision point (e.g., choice of rotation) but can occur at multiple, sometimes nested, decision points in the EFA process. In particular, nested decision points permit researchers to branch further apart from each other in response to previously selected options (i.e., the garden of forking paths). In the rotation example above, the decision between quartimin and promax rotations is nested within the decision between orthogonal or oblique rotations. In addition, consider the options for determining the number of factors. The first decision point involves the choice between PA and K1. If PA is chosen at this stage, there is a second decision point, the choice among different versions of PA (PA-F and PA-C). As we demonstrated for the case of a single replication in our simulation, results can differ substantially as researchers make different decisions at each decision point.

In addition to between-decision heterogeneity in results, the simulation also provided evidence of within-decision heterogeneity associated with a particular option. For PA-F, PA-C, and K1, we assessed the range of the suggested number of factors, which is a form of within-method heterogeneity. Of the three options, K1 showed the most extreme heterogeneity, and over-factored by as much as 19 (almost 5 times the maximum over/under for either PA method). Given the heterogeneity demonstrated within K1 (and the lack of heterogeneity within PA-F and PA-C), K1 potentially adds heterogeneity to results in two ways. First, K1 results will, on average, often differ from PA results. Second, even within the subset of researchers using K1, we would expect to see increased heterogeneity in results. For these reasons, it may be advantageous to eliminate K1 as an acceptable option.

Finally, it can be helpful to identify types of data that are particularly vulnerable to the consequences of methodological flexibility. For example, K1 almost never suggested the correct number of factors when factor loadings were weak ($\lambda = 0.45$). In such contexts, if data features are not accounted for, there is a high risk of unnecessarily adding heterogeneity to research results. For example, MacCallum et al. (1999, 2001) recommend a larger sample size for EFA when communalities are low (corroborated in the results of the current study). If a researcher is analyzing a scale with communalities that are known to

be low (many or all < 0.50 ; equivalent to $\lambda = 0.71$ which we considered “medium”), from previous research or initial estimates, then moving forward with an EFA despite only having a sample size of 60 would likely add heterogeneity to the existing literature, in addition to other known issues with small samples. Additionally, since some features of the data can be decided upon *a priori*, researchers should be encouraged to make sound decisions with respect to the data, which ameliorates some of the issues we identified that occur later in the process.

Limitations and Future Directions

One limitation of the current study is the “well-behaved” nature of the simulated data. For example, we only considered continuous data. In practice, however, it is common for data to be categorical in nature (e.g., ordinal item responses), so the results of the current study may not directly apply to item factor analysis. Data were also generated with independent clustering, no model misspecification, and the model fit to the sample was the same as the generating model. This is likely much simpler than data that would be encountered in practice. Additionally, some aspects of the simulation represented an ideal case. This included specifying the correct number of factors for EFAs and PCAs regardless of the answers given by the number of factors methods, which misses the compounding of effects from one factor analysis stage to another. However, if data were generated to be more realistic and analyses allowed for the compounding of effects, it would almost certainly lead to increases in observed heterogeneity of results.

Another limitation of the simulation design is the approach for comparing factor loading congruence between conditions using the full-length scale and conditions using a shortened version. There may be a better rule for dropping indicators that would not artificially simplify recovery and better capture the effect of shortening scales as is done in practice. Lastly, we note that the constraints of a simulation do not allow for subject matter expertise to play a role in decisions as analyses unfold. For example, the number of factors decisions here may appear worse than they would in practice to the extent that users “intervened” and used their subject matter expertise to deviate from the rote application of decision algorithms.

As an initial examination of the impact of methodological flexibility on heterogeneity in EFA, we prioritized the number of decision points and were limited in the number of options we could include within each decision point. Having identified influential EFA decision points (e.g., method for determining the number of factors and rotation) in this study, it would be beneficial for future work to compare more comprehensive sets of EFA options. For example, Next Eigenvalue Sufficiency Tests (a method for determining the number of factors; Achim, 2017) and oblimin rotation have been shown to perform well and should be included in future simulation work.

Given the observed effects of methodological choice with these somewhat simplified data, it would also be important for future research to examine the impact on factor analysis results under more realistic conditions. This study provides conservative initial estimates for how results are affected by methodological flexibility, and there is an opportunity for future research to assess the extent of impact of the issues identified in this study as well as identify the most impactful decision points or stages for factor analysis. Future research would also benefit from extending the current simulation

design to CFA. This would provide a more complete timeline of how methodological flexibility impacts heterogeneity across multiple stages of factor analytic research. More specifically, incorporating CFA into this framework would permit a direct examination of how flexibility at an earlier research stage can yield heterogeneity in results downstream. In addition, it would be valuable to extend the current work to other factor analysis models that are commonly applied in psychological research such as higher-order and bifactor models.

Conclusion

In this study, we demonstrate how methodological choices can impact heterogeneity in EFA results. These findings represent an important step toward better understanding heterogeneity in factor analysis and provide a basis for how to evaluate factor analytic replication. Given that successful replications of factor structure are crucial for ensuring that score interpretations and uses are meaningful, we hope the present work encourages greater consideration of the role of psychometrics in replication and highlights the serious implications if such considerations continue to be overlooked.

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